

Amrit Singh

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PROFESSIONAL SUMMARY

AI/ML Engineer and Web Developer with expertise in Python, Machine Learning, React.js, Node.js, Express.js, and MongoDB. Passionate about building scalable web applications and AI-powered solutions with clean, efficient code.

WORK EXPERIENCE

R&D | Internship

Nov 2025 - May 2026

Samsung

Remote

GuideWeave — AI Framework for Generating Step-by-Step Visual Guides from Knowledge Bases

- **Developed** an AI-powered platform that generates **visual repair guides** for Samsung appliances based on user-reported faults and device specifications.
- **Constructed** a multimodal RAG pipeline using LangChain, ChromaDB, Flask, and HuggingFace embeddings to effectively retrieve **instructions and diagrams** from PDF manuals.
- **Automated** document parsing and image understanding with PyMuPDF and Vision LLMs for **semantic captioning** and contextual retrieval.
- **Engineered** a responsive full-stack web application utilizing React, Vite, and Tailwind CSS, featuring optimized vector search and cached embeddings for **improved retrieval speed**.

Technologies / Skills Used : Python, Flask, React, Multimodal RAG, ChromaDB, LLMs

Web Developer | Internship

Sep 2025 - Nov 2025

Techsolid Development

Remote

- **Applied** frontend development concepts using **HTML**, **CSS**, **JavaScript**, and **Bootstrap**.
- **Built** responsive web pages and UI components utilizing the **Bootstrap** framework.
- **Employed Node.js** fundamentals for backend development and server-side programming.
- **Developed** basic **REST API** functionality and backend logic.

Technologies / Skills Used : Node.js, React, Express, MongoDB, Bootstrap

EDUCATION

International Institute of Information Technology Naya Raipur | Naya Raipur, Chhattisgarh

Aug 2023 - May 2027

B.Tech ECE (CGPA : 8.75/10)

PROJECTS

VoicePay — Voice-Based Secure Payment System (IIT Bhilai) |

Jan 2025

- Developed a secure payment platform enabling users to perform transactions using voice authentication through a Flutter-based mobile application.
- Implemented QR-code-based authentication for secure login between mobile and web platforms for net banking payments.
- Built secure user authentication and device verification workflows to enhance transaction security and user accessibility.
- Designed and integrated mobile-to-web communication features for seamless cross-platform payment authorization.

Technologies / Tools Used : React, Node.js, Supabase, LLMs

AI-Driven Adaptive Learning Platform : ModuLearn

Jan 2026

- Developed an **AI-powered learning platform** that generates personalized study materials from various sources, including **topics**, **PDFs**, and **web content**.
- Implemented features such as **PDF summarization**, **AI-generated explanations**, quizzes, flashcards, and **YouTube video recommendations** using LLMs.
- Designed adaptive learning workflows and interactive tutoring features to enhance user engagement.
- Utilized the **Manim** Python library to create educational videos.

Technologies / Tools Used : React, Node.js, LLM, Supabase, Data preprocessing, Manim

SKILLS

AI / Machine Learning :

Pandas, Data Analysis, LangChain, RAG, LLM, NLP, Deep Learning

Frameworks & Libraries :

React, Next.js, Tailwind CSS, FastAPI

Programming Languages :

C++, Java, JavaScript, SQL, Python

Databases :

PostgreSQL, MySQL, MongoDB, Supabase

Tools & Platforms :

Git, Postman, CI/CD, Docker

Languages :

English, Hindi